

**SEPTEMBER 2026 UPDATE**  
**ANNA SOFIA MORO**

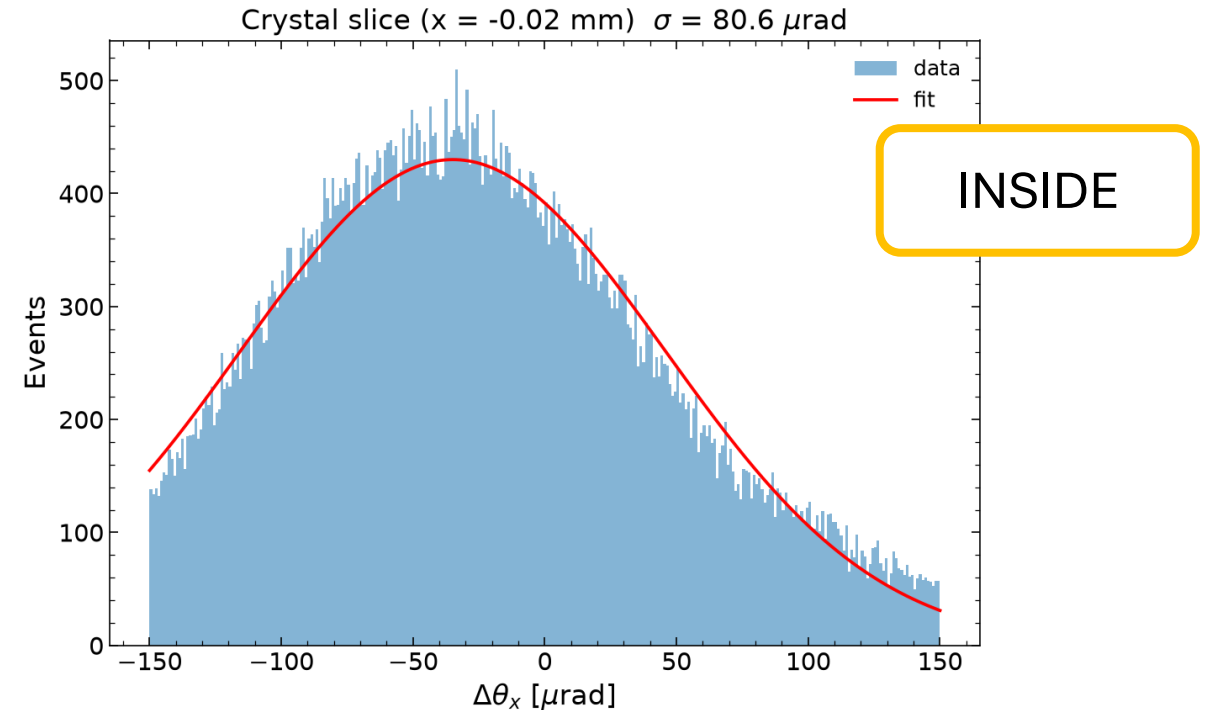
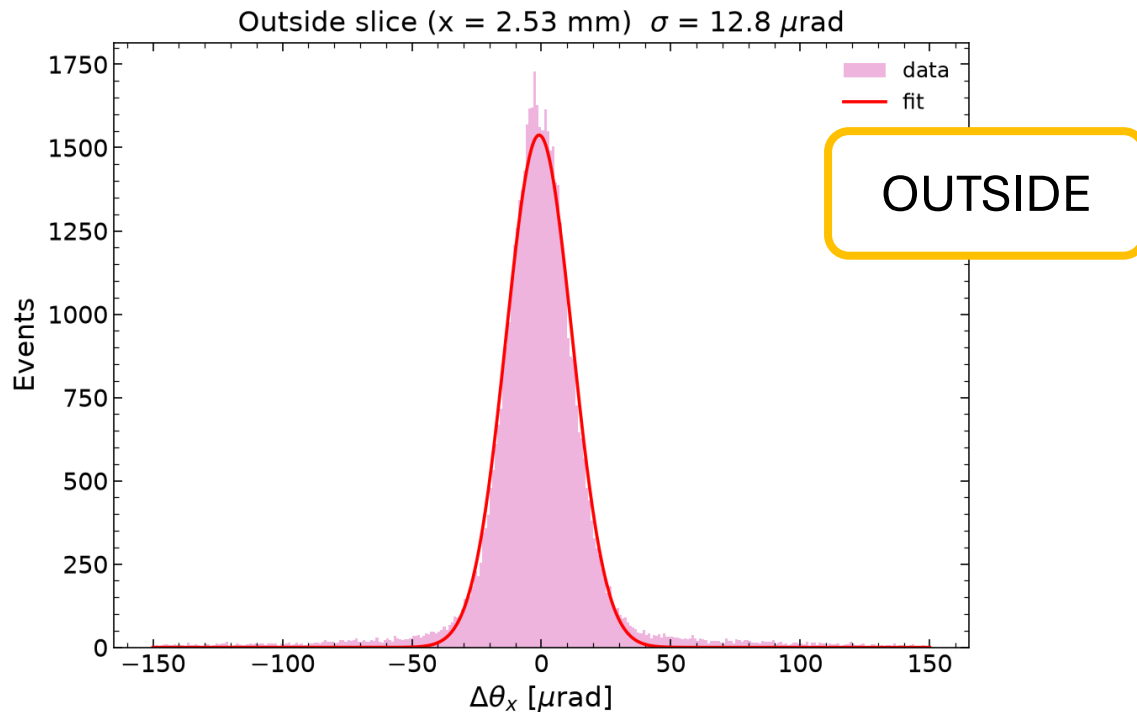
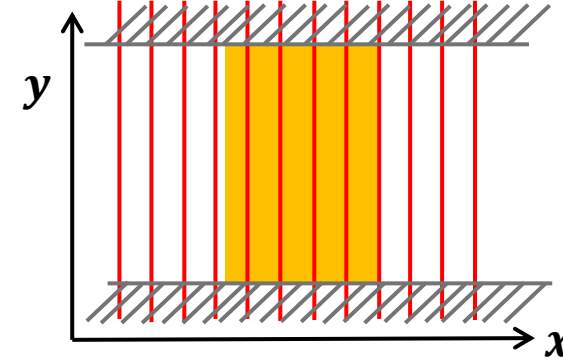
# WHAT'S NEW?

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1. Multiple Coulomb Scattering
  - Analysis made for both x and y
  - Clamp material evaluation
2. Analysis of the beam distribution of real data
  - Check the distributions
  - Check the correlation/covariance
3. Simulation
  - Reproduce the real data beam features
  - Channeling efficiency analysis
4. Thesis
  - Possible order of topics in the table of contents

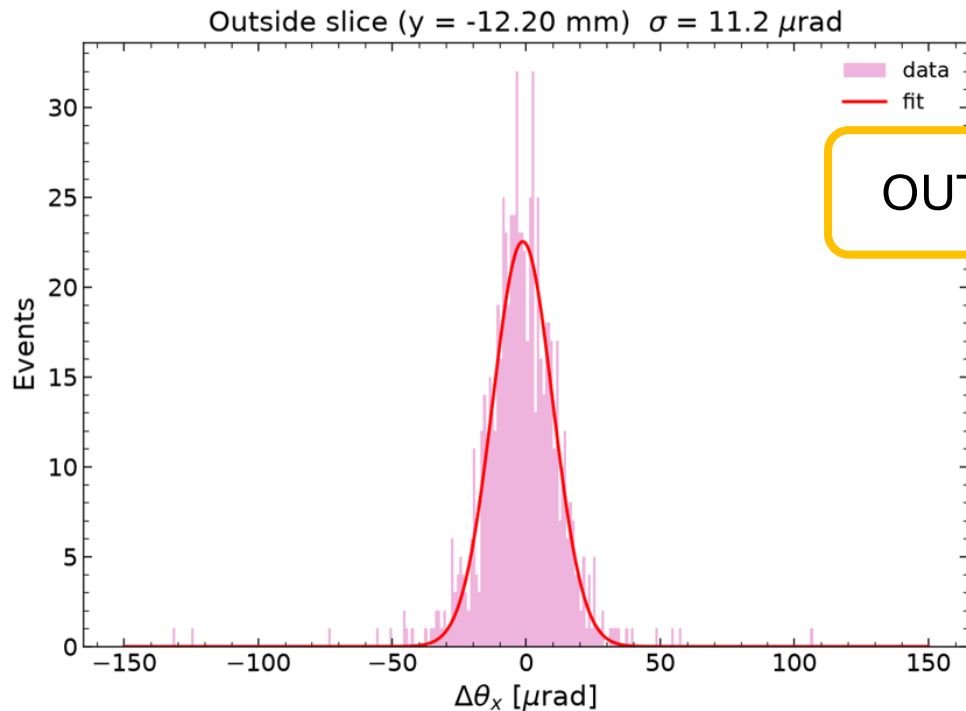
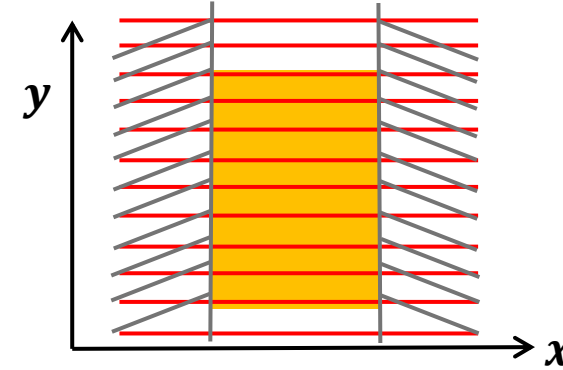
# FIND THE CRYSTAL FOOTPRINT THROUGH SCATTERING

- Divided the crystal into slices along  $x$  or  $y$
- Width of the angular distribution  $\sigma$  for each slice

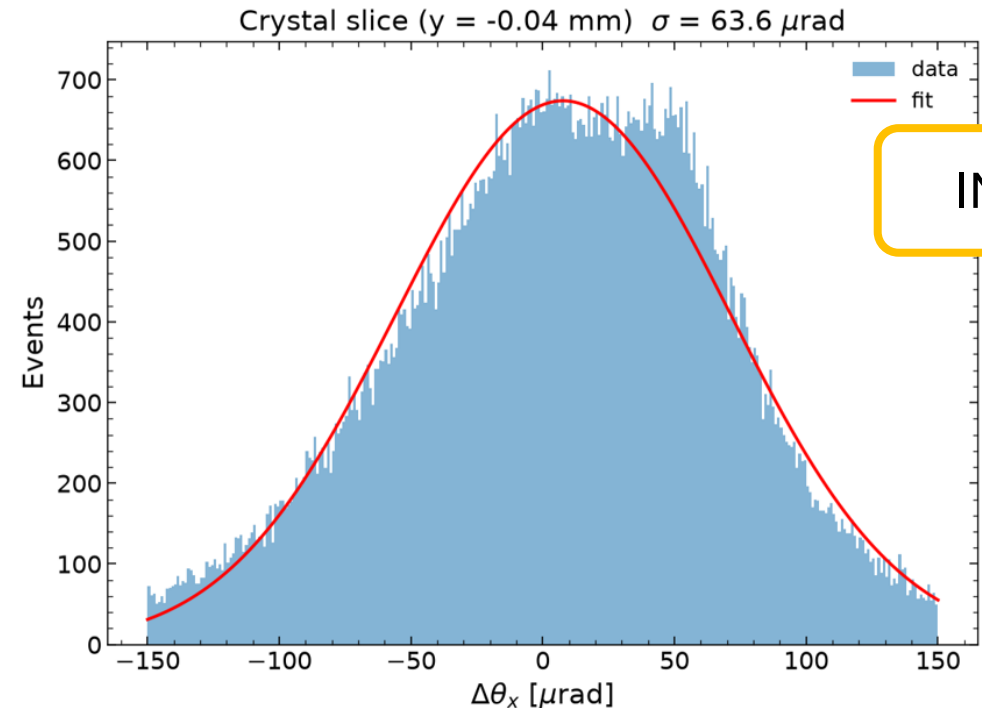


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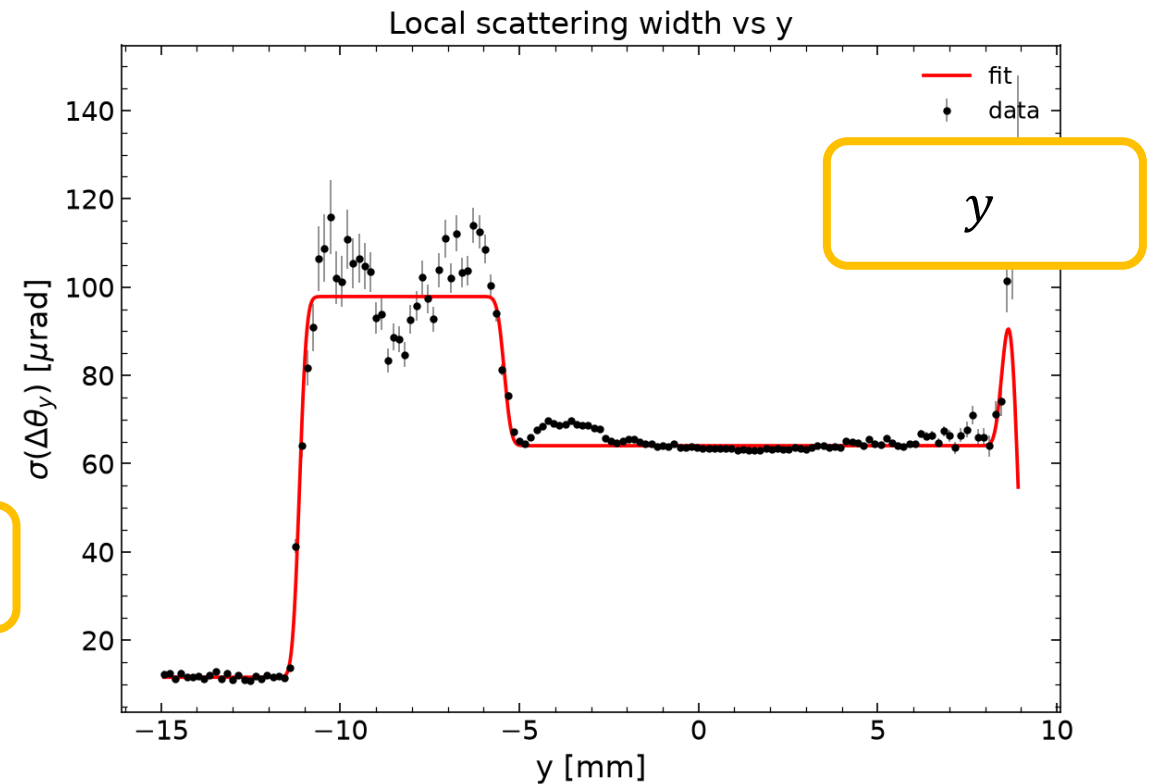
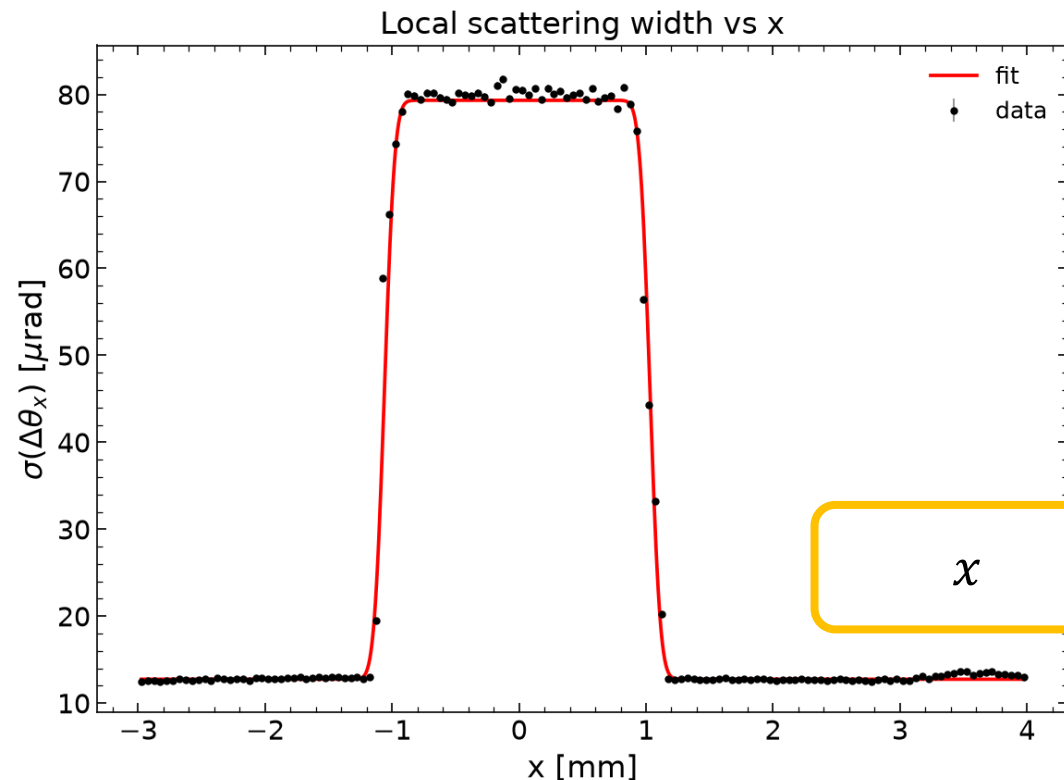
OUTSIDE



INSIDE

# FIND THE CRYSTAL FOOTPRINT THROUGH SCATTERING

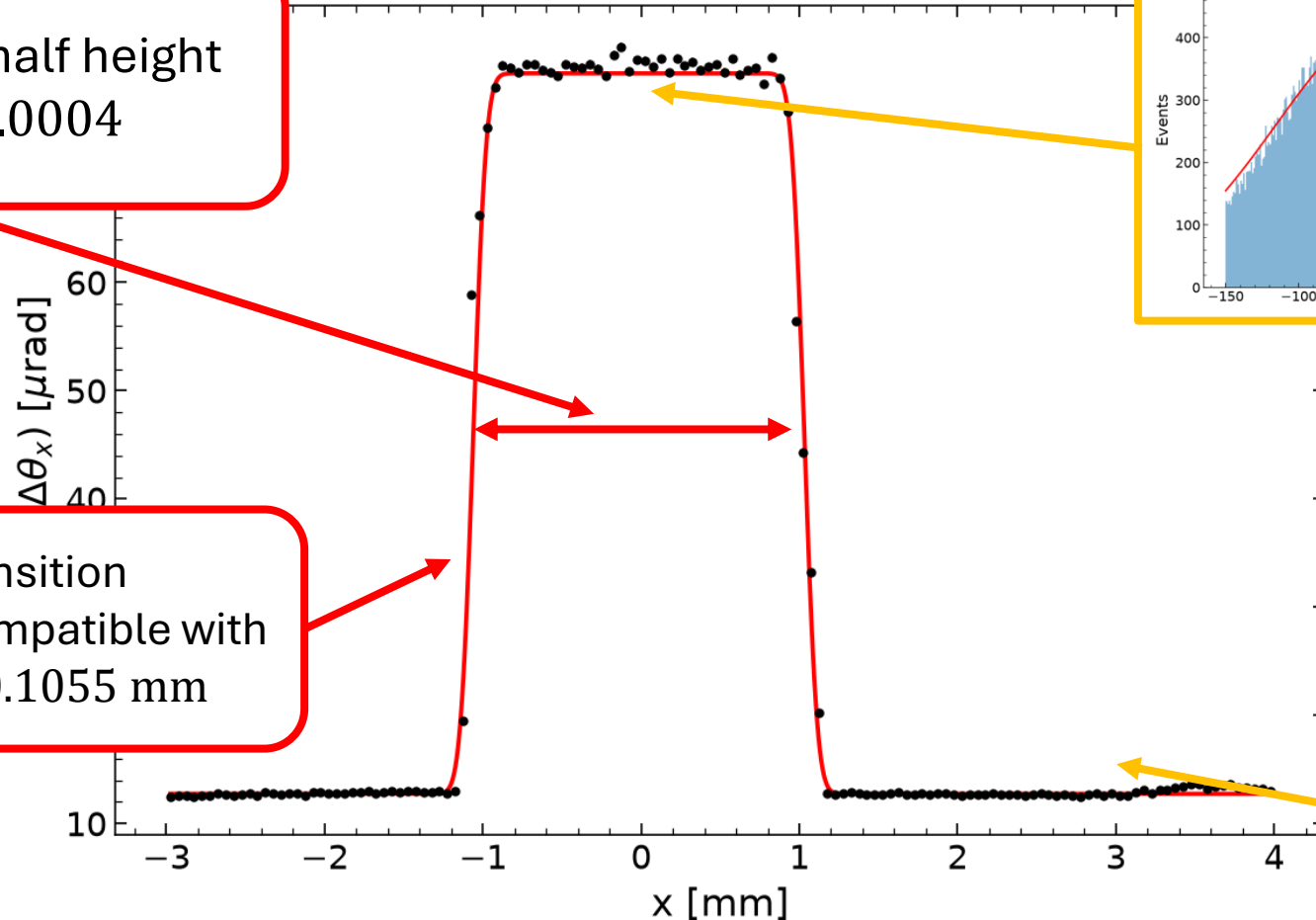
- The extracted gaussian widths  $\sigma$  are plotted against the impact position
- The transition between the outside and inside regions is fitted using a Error function (Erf)



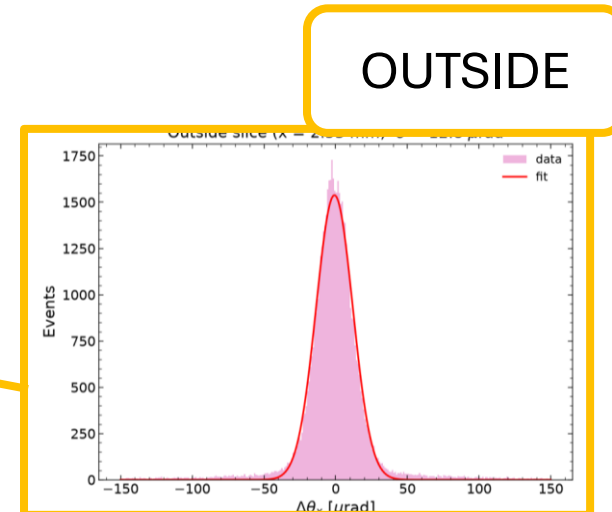
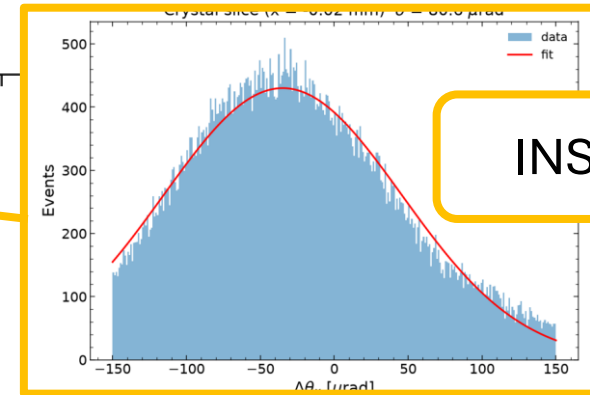
# FIND THE CRYSTAL FOOTPRINT THROUGH SCATTERING

The window @ half height  
 $2.0911 \pm 0.0004$

Local scattering width vs x

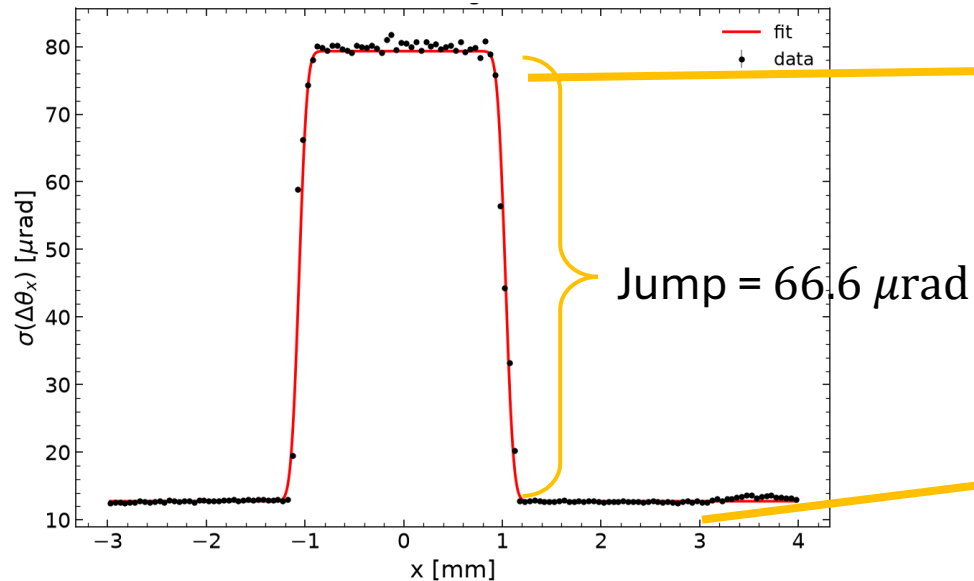


The width transition  
 $0.0901(3)$  mm is compatible with  
position error  $\sim 0.1055$  mm



# FIND THE CRYSTAL FOOTPRINT THROUGH SCATTERING

$$\langle \theta_{MCS} \rangle = \frac{13.6 \text{ MeV}}{\beta c p} z \sqrt{\frac{x}{X_0}} \left[ 1 + 0.038 \ln \left( \frac{x z^2}{X_0 \beta^2} \right) \right] = 66.5 \mu\text{rad}$$



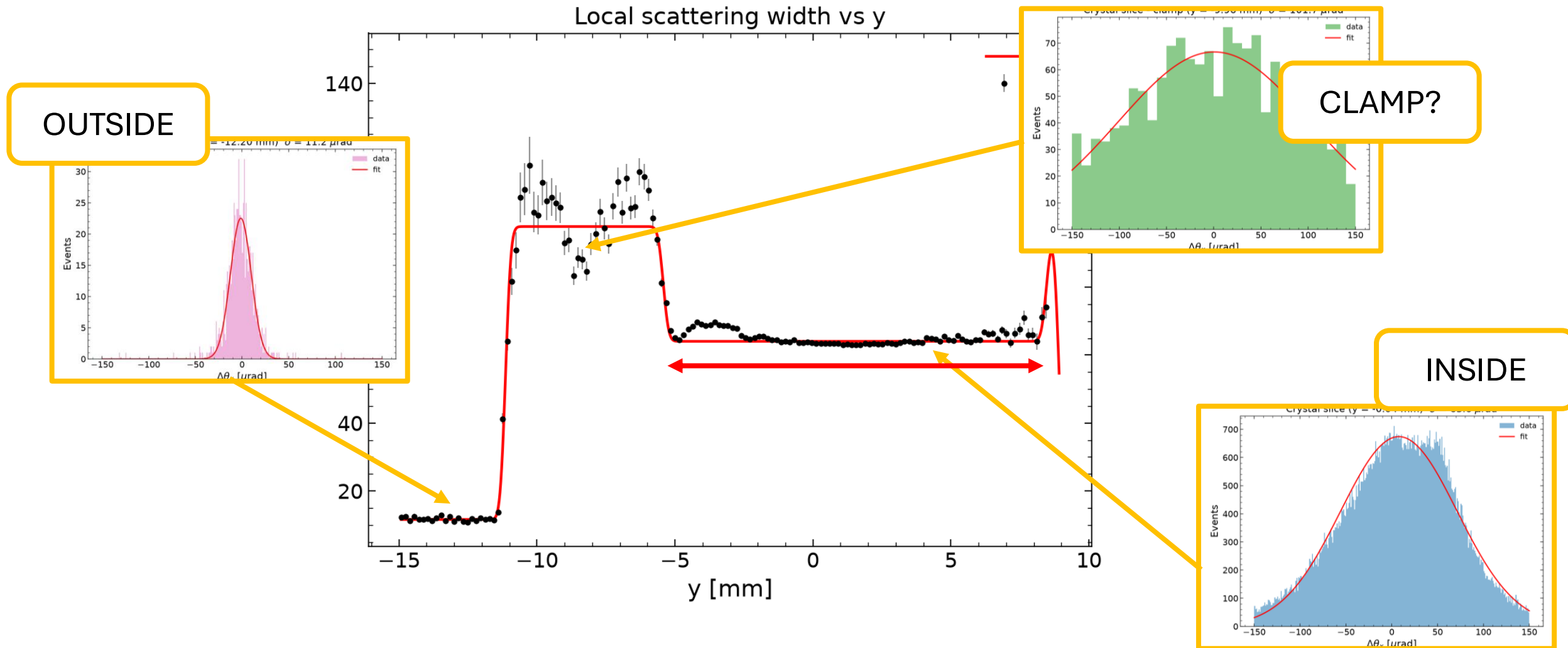
$$\sigma_{meas} = \sqrt{\sigma_{mcs}^2 + \sigma_{track}^2} \sim 79.5 \mu\text{rad}$$

$$\rightarrow \sigma_{mcs} = 78.30(6) \mu\text{rad} \neq \sigma_{mcs}^{th}$$

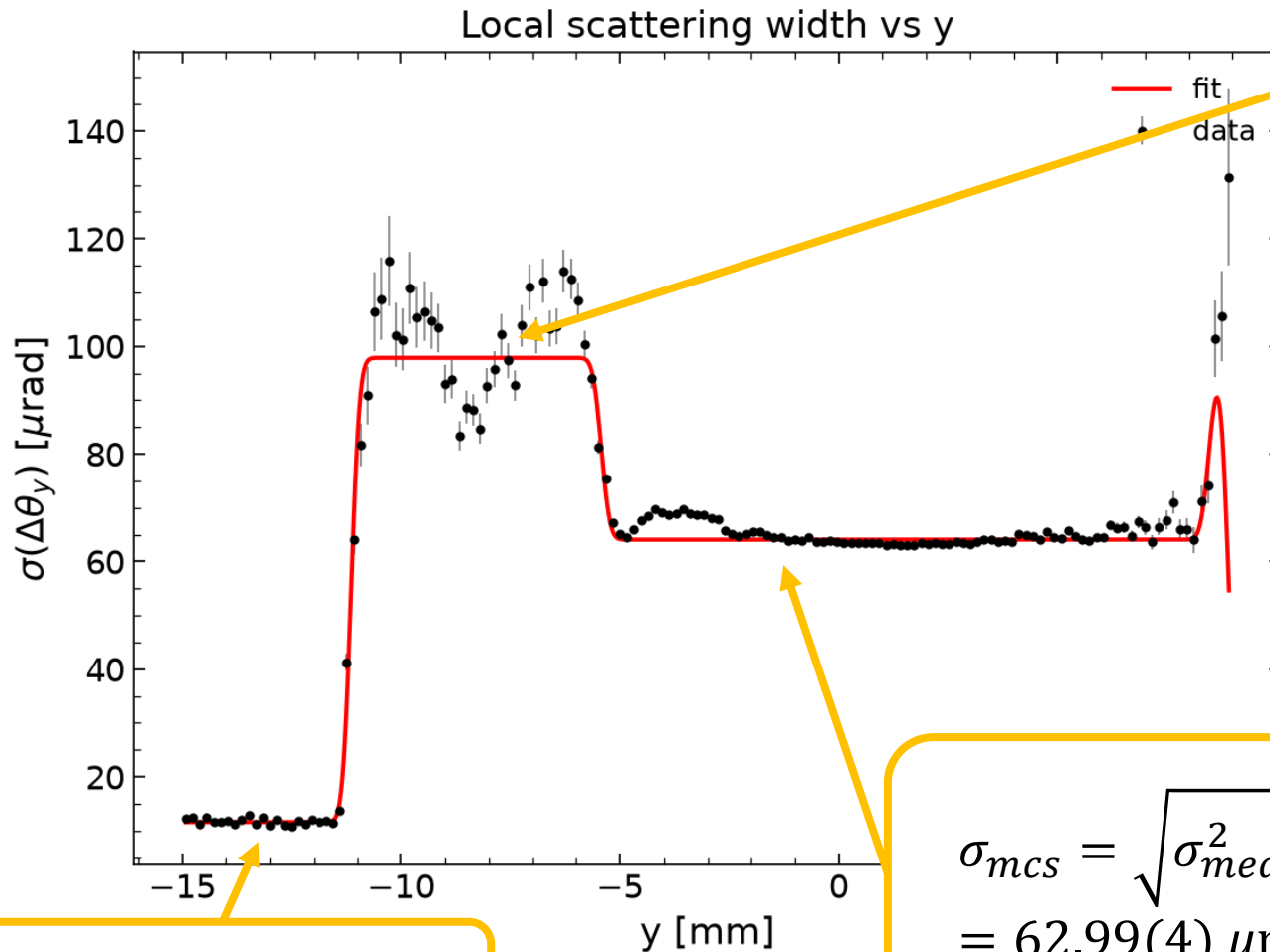
not zero, but  $\sim 12.75(1) \mu\text{rad}$   
which corresponds to  $\sigma_{track}$

$$\sigma_{track} = \sqrt{\sigma_{track,in}^2 + \sigma_{track,out}^2} = 12.6 \mu\text{rad}$$

# FIND THE CRYSTAL FOOTPRINT THROUGH SCATTERING



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CLAMP: which material/metal?  
on average  $\sim 99.2 \pm 0.3 \mu\text{rad}$

*unknown:  $x, X_0$*

Between different metallic materials (stainless steel, aluminium, titanium, copper, brass):

→ The only similar is titanium

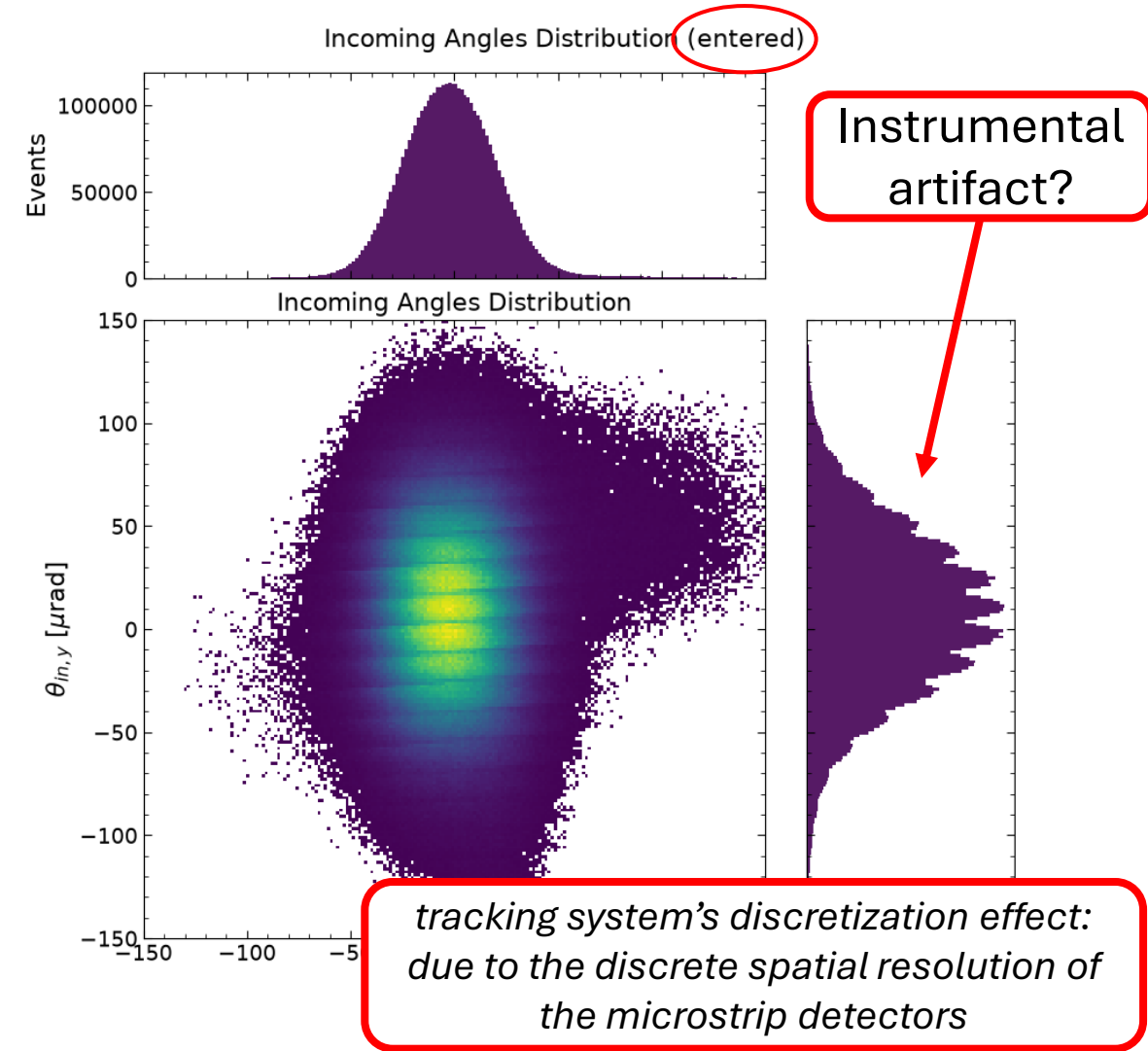
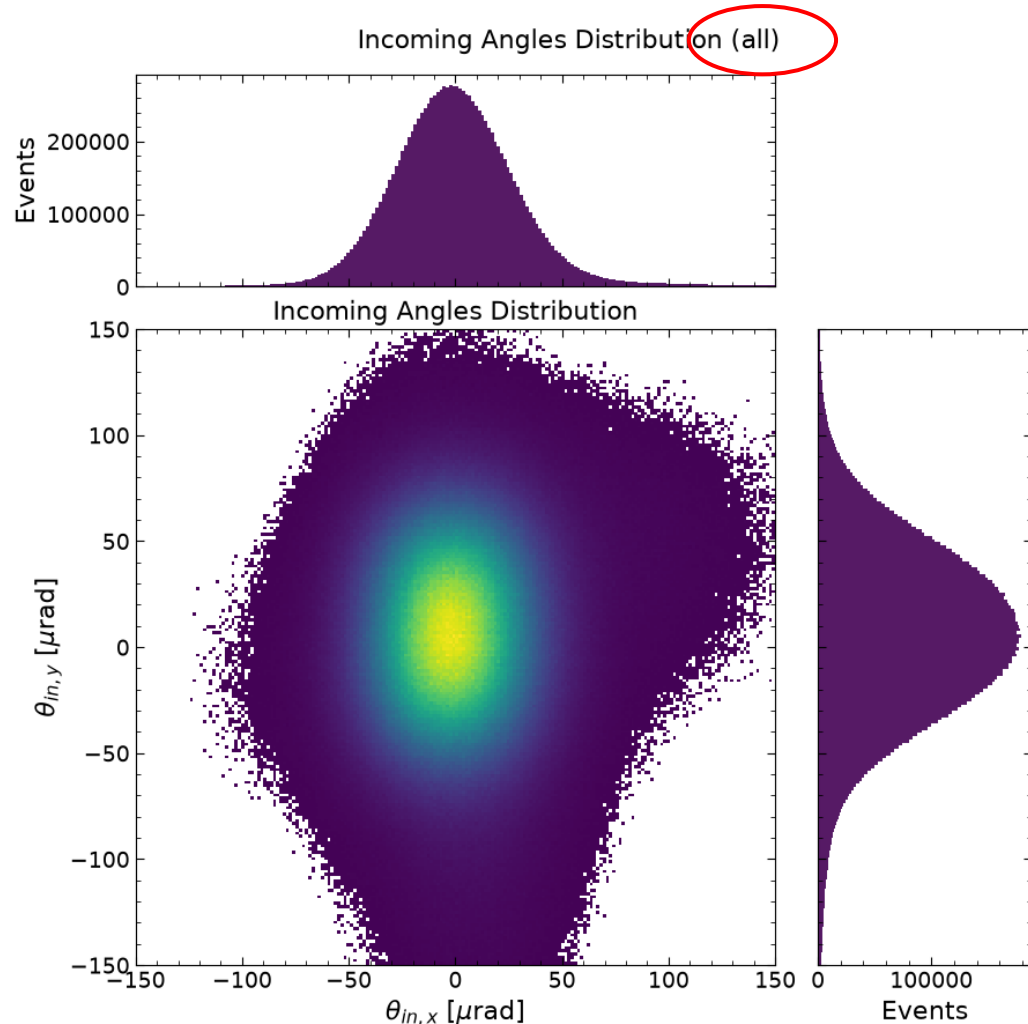
$\theta \sim 110 \mu\text{rad}$

*(stainless steel too much scattering, aluminium not enough scattering)*

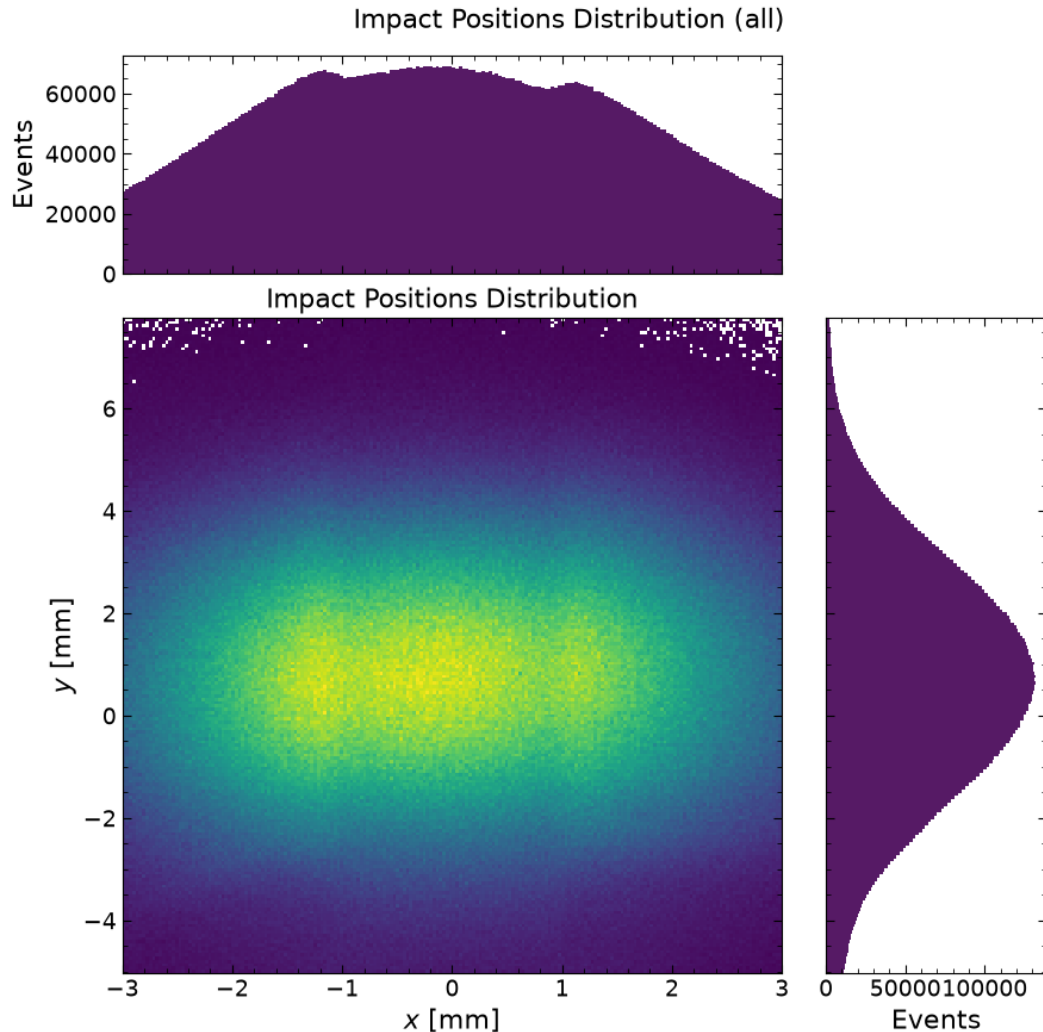
$$\sigma_{bsl} = 11.6(1) \mu\text{rad}$$

$$\sigma_{mcs} = \sqrt{\sigma_{meas}^2 - \sigma_{bsl}^2}$$
$$= 62.99(4) \mu\text{rad}$$

# ANALYSIS OF THE BEAM DISTRIBUTION (REAL DATA)



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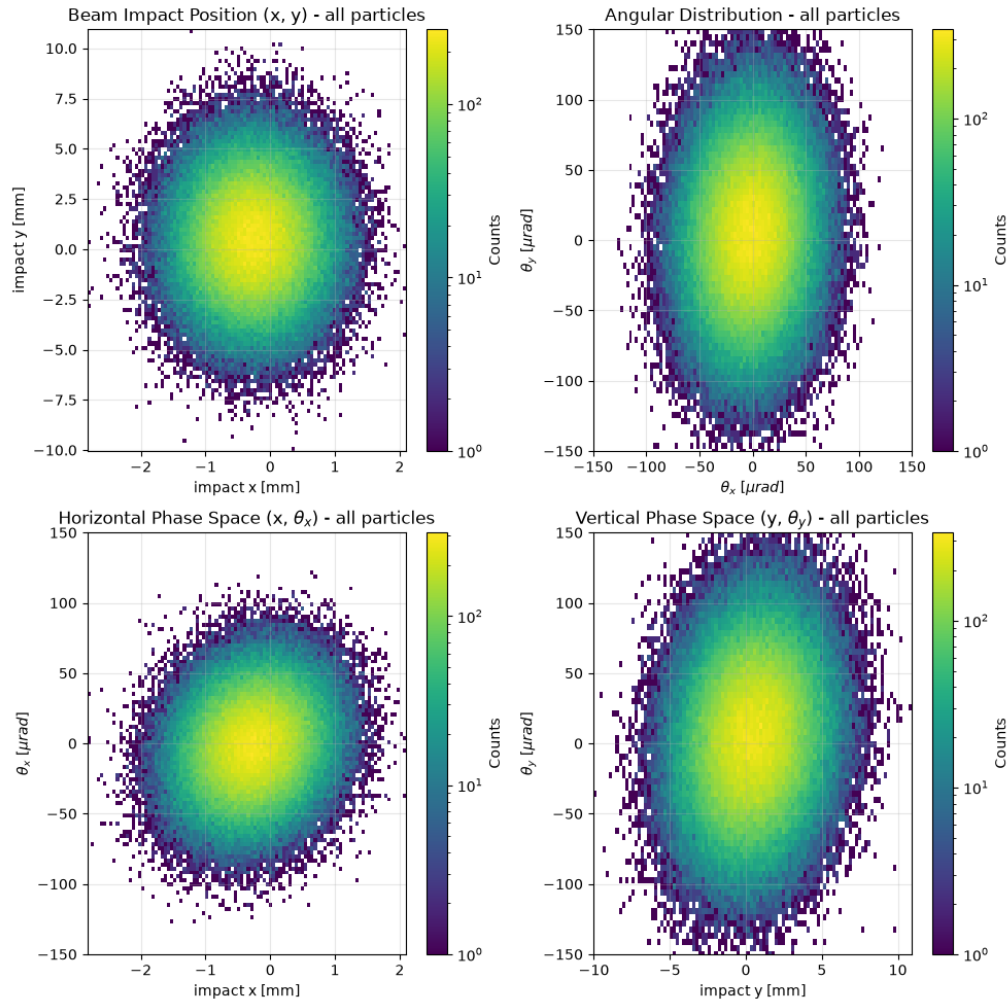
✓ Gaussian fits on the distributions of  $x, y, \theta_x, \theta_y$

✓ Check the correlation:

Not negligible for  $x - \theta_x, y - \theta_y$ :

|            | $x$ | $y$ | $\theta_x$ | $\theta_y$ |
|------------|-----|-----|------------|------------|
| $x$        | 1   | 0   | 0.2        | 0          |
| $y$        | 0   | 1   | 0          | 0.2        |
| $\theta_x$ | 0.2 | 0   | 1          | 0          |
| $\theta_y$ | 0   | 0.2 | 0          | 1          |

# SIMULATING THE SHORT CRYSTAL



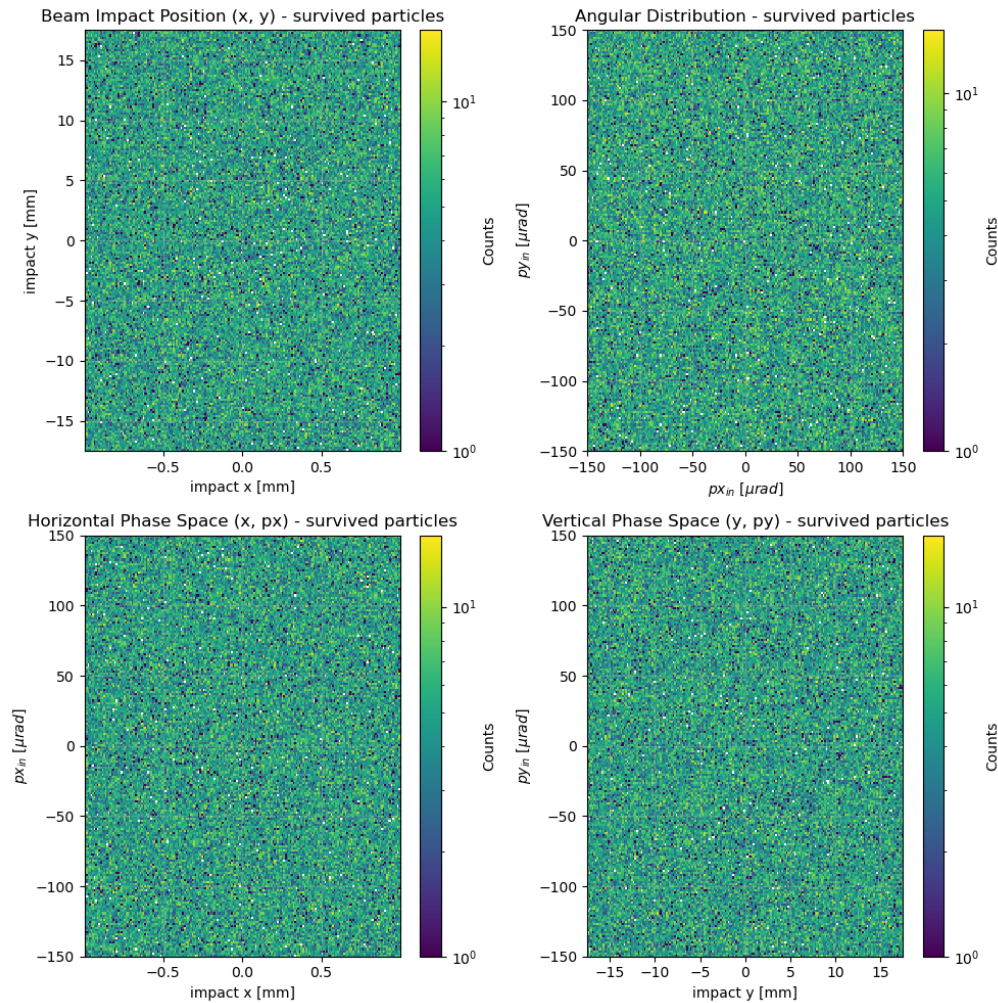
Fitting the 1d marginal distributions and then sampling them independently would not be correct if variables are uncorrelated.

**Solution:**

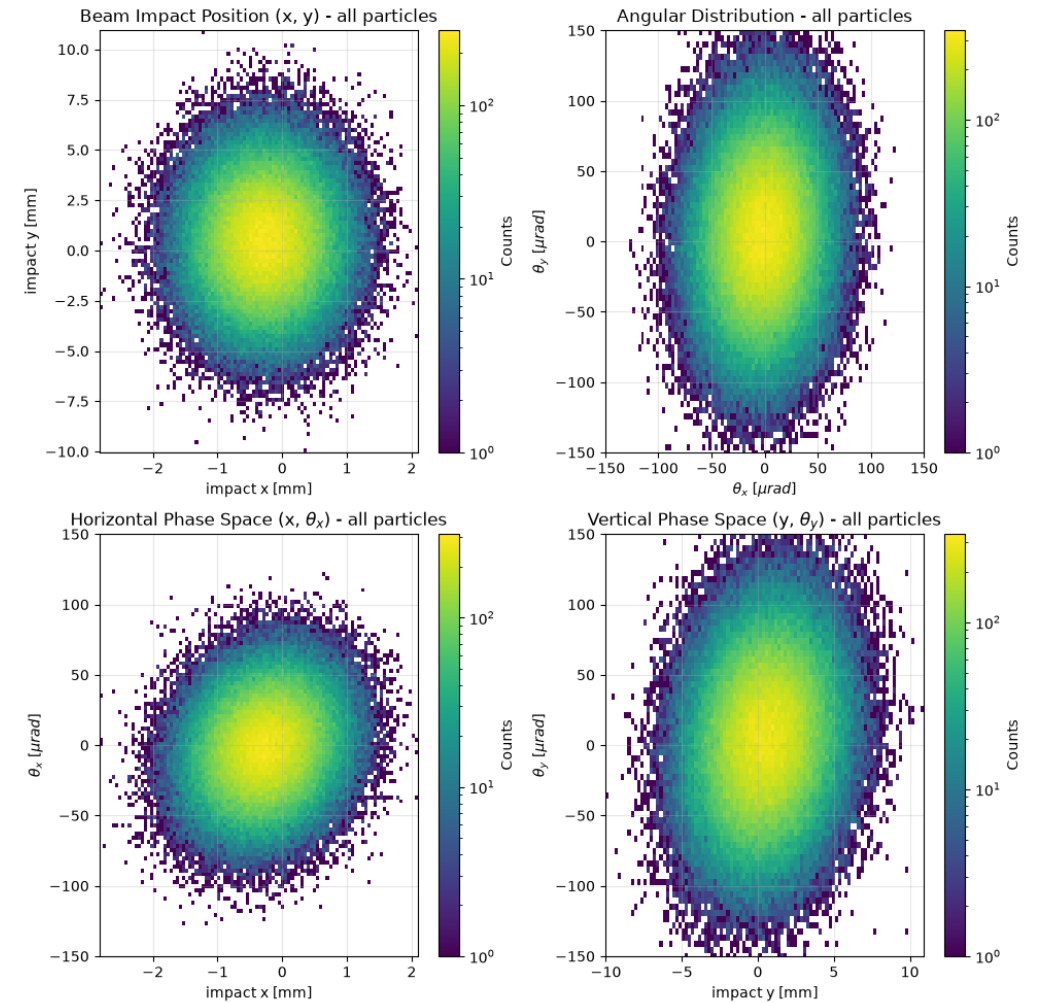
Using a multivariate random normal distribution (thanks to the covariance matrix reproduce the correlation)

# SIMULATING THE SHORT CRYSTAL

Before: normal distribution

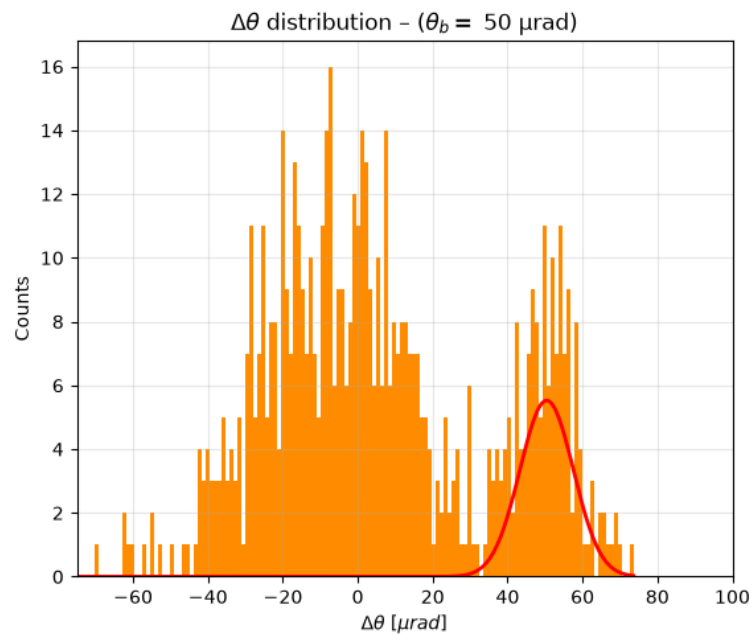


Now: normal distribution



# SIMULATING THE SHORT CRYSTAL

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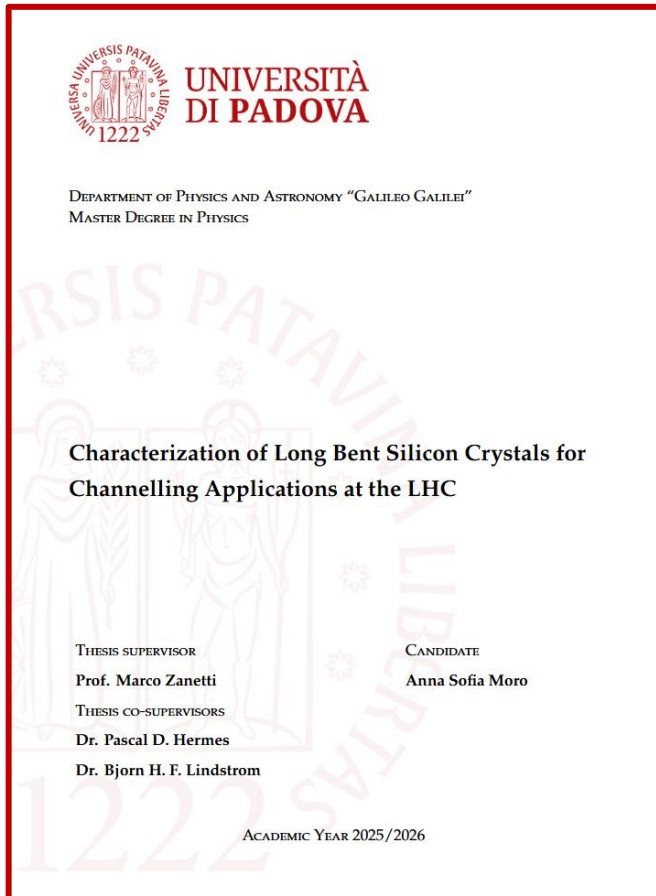
Out of 200k particles only  $\sim 650$  are surviving

But with  $\sigma_x = 0.568$  mm and the crystal with 1 mm width, the fraction that should hit the crystal is 92%.

Possible problems?

# THESIS: PART I

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- Physics background
  - Dipole moment
  - $\Lambda_C^+$
- Crystal Channeling
  - In straight/bent crystals
  - Other processes (dechannelling, volume capture, ...)
- Experimental principle: measuring spin precession through bent crystals
- TWOCRIST experiment
- H8 testing